

torch.signal.windows.general_cosine#

https://pytorch.org/docs/stable/generated/torch.signal.windows.general_cosine.html

Description:

Computes the general cosine window. The general cosine window is defined as follows: The window is normalized to 1 (maximum value is 1). However, the 1 doesn't appear if M is even and sym is True. M (int) – the length of the window. In other words, the number of points of the returned window. a (Iterable) – the coefficients associated to each of the cosine functions.

Function Signature

```
torch.signal.windows.general_cosine(M, *, a, sym=True,
dtype=None, layout=torch.strided, device=None,
requires_grad=False)[source]#
```

Parameters

Keyword Arguments

Return type

Parameters

Keyword

`a` (Iterable) – the coefficients associated to each of the cosine functions. `sym` (bool, optional) – If False, returns a periodic window suitable for use in spectral analysis. If True, returns a symmetric window suitable for use in filter design. Default: True. `dtype` (torch.dtype, optional) – the desired data type of returned tensor. Default: if None, uses a global default (see `torch.set_default_dtype()`). `layout` (torch.layout, optional) – the desired layout of returned Tensor. Default: `torch.strided`. `device` (torch.device, optional) – the desired device of returned tensor. Default: if None, uses the current device for the default tensor type (see `torch.set_default_device()`). device will be the CPU for CPU tensor types and the current CUDA device for CUDA tensor types. `requires_grad` (bool, opt

Returns:

Tensor

Code Examples

```
>>> # Generates a symmetric general cosine window with 3
coefficients.
>>> torch.signal.windows.general_cosine(10, a=[0.46, 0.23,
0.31], sym=True)
tensor([0.5400, 0.3376, 0.1288, 0.4200, 0.9136, 0.9136, 0.4200,
0.1288, 0.3376, 0.5400])

>>> # Generates a periodic general cosine window with 2
coefficients.
>>> torch.signal.windows.general_cosine(10, a=[0.5, 1 - 0.5],
sym=False)
tensor([0.0000, 0.0955, 0.3455, 0.6545, 0.9045, 1.0000, 0.9045,
0.6545, 0.3455, 0.0955])
```

Detailed Documentation

Documentation

Computes the general cosine window.

The general cosine window is defined as follows:

The window is normalized to 1 (maximum value is 1). However, the 1 doesn't appear if M is even and `sym` is `True`.

M (int) – the length of the window. In other words, the number of points of the returned window.

`a` (Iterable) – the coefficients associated to each of the cosine functions.

`sym` (bool, optional) – If False, returns a periodic window suitable for use in spectral analysis. If True, returns a symmetric window suitable for use in filter design. Default: True.

`dtype` (torch.dtype, optional) – the desired data type of returned tensor. Default: if None, uses a global default (see `torch.set_default_dtype()`).

`layout` (torch.layout, optional) – the desired layout of returned Tensor. Default: `torch.strided`.

`device` (torch.device, optional) – the desired device of returned tensor. Default: if None, uses the current device for the default tensor type (see `torch.set_default_device()`). device will be the CPU for CPU tensor types and the current CUDA device for CUDA tensor types.

`requires_grad` (bool, optional) – If autograd should record operations on the returned tensor. Default: False.